Percentages and Proportion

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS percentage *(ōrau)* · per cent *(ōrau)* · of · the whole (100%) · for every · proportion *(ōwehenga ōrite)*

A Common percentages ●○○

Remember: 50% is a half, 25% is a quarter, 10% is a tenth and 1% is a hundredth. So 10% of 70 is $70 \div 10 = 7$.

1. Work out.

a) 50% of 60 = _____ **b)** 25% of 80 = _____ **c)** 10% of 70 = _____ **d)** 1% of 300 = _____

e) 50% of 18 = _____ **f)** 25% of 40 = _____ **g)** 10% of 250 = _____ **h)** 1% of 800 = _____

i) 20% of 45 = _____ **j)** 75% of 40 = _____ **k)** 5% of 120 = _____ **l)** 10% of 65 = _____

m) 50% of 37 = _____ **n)** 25% of 10 = _____ **o)** 100% of 46 = _____ **p)** 5% of 60 = _____

B Building other percentages ●●○

Remember: Find 10% first, then multiply. For 30% of 60: 10% is 6, so 30% is $6 \times 3 = 18$.

2. Work out.

a) 30% of 60 **b)** 70% of 40 **c)** 40% of 25 **d)** 90% of 150

e) 60% of 45 **f)** 20% of 85 **g)** 30% of 200 **h)** 80% of 15

3. Work out.

a) 10% of \$45 **b)** 25% of \$18 **c)** 50% of 250 g **d)** 20% of 40 kg

C Finding the whole ●●●

Watch out: Do not find the percentage of the number you are given. Work back to 100%: if 20% of a number is 8, then 100% is $8 \times 5 = 40$.

4. Find the number.

a) 50% of a number is 14 **b)** 10% of a number is 7 **c)** 25% of a number is 9

d) 20% of a number is 8 **e)** 5% of a number is 3 **f)** 1% of a number is 2

g) 50% of a number is 2.5 **h)** 10% of a number is 12 **i)** 25% of a number is 20

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D "For every" problems ●●●

> **Remember:** Work out how many groups there are, then multiply. "For every 2 red beads there are 3 blue beads" and 10 red beads means 5 groups, so $5 \times 3 = 15$ blue beads.

5. Answer each question.

a) For every 2 red beads on a necklace there are 3 blue beads. There are 10 red beads. How many blue beads are there?

b) For every 4 students on a trip there is 1 adult. There are 28 students. How many adults are there?

c) A recipe uses 3 cups of flour for every 2 eggs. How many eggs are needed for 15 cups of flour?

d) A shop gives 2 free pens for every 5 pens bought. A school buys 40 pens. How many free pens does it get?

e) For every $10 Tane earns, he saves $3. He earns $60. How much does he save?

E Problems ●●●

6. A test has 36 questions. Kiri gets 50% of them right. How many questions does she get right?

7. Sam saves 20% of the $60 he earns. How much does he save?

8. 15 students in a class walk to school. This is 50% of the class. How many students are in the class?

★ Extension

9. Find 15% of 80 by adding 10% and 5%.

10. For every 2 girls in a club there are 3 boys. There are 30 children altogether. How many girls are there?

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Percentages and Proportion

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS percentage *(ōrau)* · per cent *(ōrau)* · of · the whole (100%) · for every · proportion *(ōwehenga ōrite)*

A Common percentages ●○○

Remember: 50% is a half, 25% is a quarter, 10% is a tenth and 1% is a hundredth. So 10% of 70 is $70 \div 10 = 7$.

1. Work out.

a) 50% of 80 = _____ **b)** 25% of 60 = _____ **c)** 10% of 90 = _____ **d)** 1% of 500 = _____

e) 50% of 14 = _____ **f)** 25% of 20 = _____ **g)** 10% of 350 = _____ **h)** 1% of 600 = _____

i) 20% of 35 = _____ **j)** 75% of 80 = _____ **k)** 5% of 140 = _____ **l)** 10% of 45 = _____

m) 50% of 29 = _____ **n)** 25% of 30 = _____ **o)** 100% of 52 = _____ **p)** 5% of 80 = _____

B Building other percentages ●●○

Remember: Find 10% first, then multiply. For 30% of 60: 10% is 6, so 30% is $6 \times 3 = 18$.

2. Work out.

a) 30% of 80 **b)** 70% of 60 **c)** 40% of 35 **d)** 90% of 250

e) 60% of 35 **f)** 20% of 65 **g)** 30% of 400 **h)** 80% of 25

3. Work out.

a) 10% of $35 **b)** 25% of $14 **c)** 50% of 350 g **d)** 20% of 60 kg

C Finding the whole ●●●

Watch out: Do not find the percentage of the number you are given. Work back to 100%: if 20% of a number is 8, then 100% is $8 \times 5 = 40$.

4. Find the number.

a) 50% of a number is 17 **b)** 10% of a number is 9 **c)** 25% of a number is 7

d) 20% of a number is 12 **e)** 5% of a number is 4 **f)** 1% of a number is 3

g) 50% of a number is 3.5 **h)** 10% of a number is 15 **i)** 25% of a number is 30

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D "For every" problems ●●●

> **Remember:** Work out how many groups there are, then multiply. "For every 2 red beads there are 3 blue beads" and 10 red beads means 5 groups, so $5 \times 3 = 15$ blue beads.

5. Answer each question.

a) For every 2 red beads on a necklace there are 5 blue beads. There are 8 red beads. How many blue beads are there?

b) For every 5 students on a trip there is 1 adult. There are 35 students. How many adults are there?

c) A recipe uses 4 cups of flour for every 3 eggs. How many eggs are needed for 20 cups of flour?

d) A shop gives 2 free pens for every 3 pens bought. A school buys 24 pens. How many free pens does it get?

e) For every $10 Tane earns, he saves $7. He earns $50. How much does he save?

E Problems ●●●

6. A test has 36 questions. Kiri gets 25% of them right. How many questions does she get right?

7. Sam saves 10% of the $80 he earns. How much does he save?

8. 18 students in a class walk to school. This is 50% of the class. How many students are in the class?

★ Extension

9. Find 15% of 60 by adding 10% and 5%.

10. For every 3 girls in a club there are 5 boys. There are 40 children altogether. How many girls are there?

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